

VARIABLES & DISPLAYS

new 1.1–1.6, 1.8–1.9 • old 1.1–1.6, 1.8–1.9

ZONE A • THE DISPLAYS

CATEGORICAL

Labels or groups

Bar graph • pie chart

Frequency table • relative frequency

GETTING TO SCHOOL

Mode	Frequency
Walk	2
Bus	4
Car	1

Frequency = count.
Relative frequency = share of the total.

QUANTITATIVE

Counts or measurements

Dotplot • stemplot • histogram

Boxplot • cumulative graph

Frequency

Commute time (minutes)

A boxplot summarizes position and spread

SHAPE • CENTER • SPREAD • OUTLIERS

symmetric • skewed left • skewed right
unimodal • bimodal • uniform

ZONE B • WHAT THE WORDS MEAN

WHO AND WHAT?

Individual: the person or object described.
Variable: a characteristic recorded for each individual.

TWO KINDS OF QUANTITATIVE DATA

Discrete: separate, countable values.
Continuous: any value in an interval.

ZONE C • SAY IT IN WORDS

“Distribution A is [more/less] _____ than B, because _____.”

Use a shape, center, or spread comparison. Support it with numbers, units, and the variable in context.

ZONE D • WATCH OUT

NAME THE TAIL

Skewed **left** means the tail is on the left, not the pile.

BINS OR CATEGORIES?

Adjacent histogram bins touch; category bars have gaps.
An empty numeric bin can still leave blank space.

SUMMARY STATISTICS

new 1.7–1.8 • old 1.7–1.8

ZONE A · THE FORMULAS

$$\bar{x} = \frac{\sum x_i}{n}$$

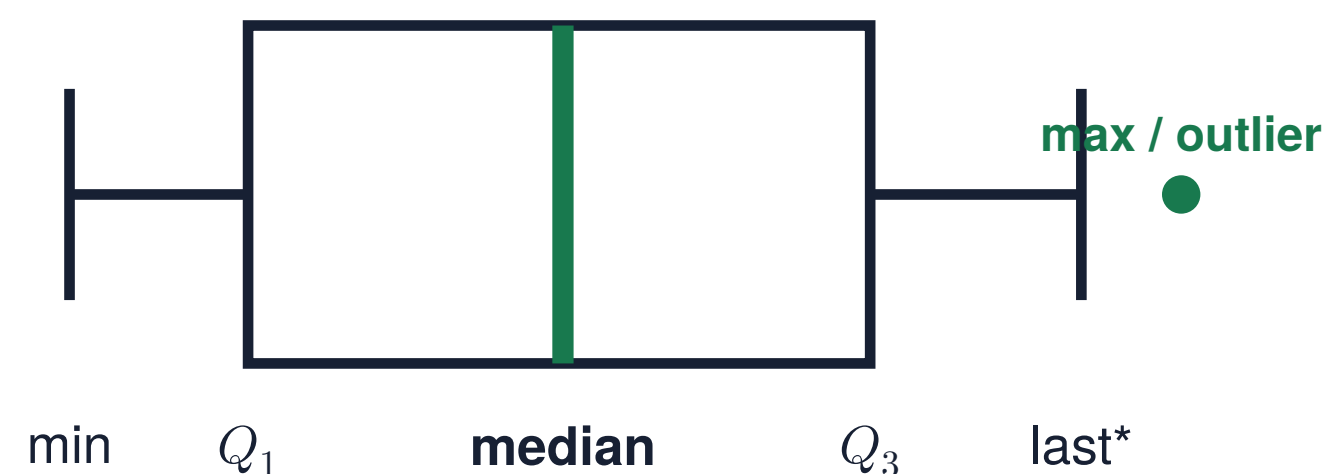
$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

$$\text{IQR} = Q_3 - Q_1$$

outlier if $x > Q_3 + 1.5(\text{IQR})$
or $x < Q_1 - 1.5(\text{IQR})$

A BOX PLOT IS A FIVE-NUMBER STORY



*The whisker stops at the last non-outlier. The maximum here is the separate point.

CENTER + SPREAD WORK AS A PAIR

symmetric, no outliers $\rightarrow \bar{x}$ and s
skewed or outliers $\rightarrow$ median and IQR

ZONE B · WHAT THE SYMBOLS MEAN

SAMPLE

$\bar{x}$ = sample mean s = sample SD
five-number summary: min, Q_1 , median, Q_3 , max

POPULATION + RESISTANCE

μ = population mean	σ = population SD
resistant: median, IQR	not resistant: mean, SD,
range	

ZONE C . SAY IT IN WORDS

“Typically, _____ varies from the mean by about _____.”

variable in context

value of s

units

ZONE D · WATCH OUT

THE DENOMINATOR

For a **sample** variance or SD, divide by $n - 1$, not n .

TRANSFORMATIONS

Adding a constant shifts the center but not the spread. Multiplying by a constant scales **both**.

COLLECTING DATA & SCOPE OF INFERENCE

new 1.10–1.13 • old 3.1–3.5 • ★ old 3.6

ZONE A • HOW THE DATA ARE COLLECTED

Data from random sample or randomized experiment

SAMPLING METHODS SRS: every possible sample of size n has the same chance. Stratified: divide into similar groups; randomly sample within each. Cluster: randomly choose whole groups; include everyone in them. Systematic: start randomly, then select every k th individual.	SOURCES OF BIAS Undercoverage: some groups are left out or hard to reach. Nonresponse: selected people do not respond. Response bias: wording or reporting distorts answers. Voluntary response: people choose to join the sample. Convenience: choose people who are easiest to reach.	EXPERIMENTS Control: hold other conditions alike; compare treatments. Random assignment: use chance to form treatment groups. Replication: use enough experimental units. Completely randomized: assign all units at random. Randomized block: group similar units, then randomize within. Matched pairs: compare two related measurements or units. Placebo: an inactive treatment. Blinding: hide treatment identity from subjects or evaluators. Confounding: effects are mixed and cannot be separated.
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ZONE B • WHAT THE WORDS MEAN

THE STUDY Observational: record what happens. Experiment: impose a treatment.	THE PEOPLE OR OBJECTS Population: the full group of interest. Sample: the group studied.	THE MEASUREMENTS Treatment: a condition imposed. Response variable: the outcome measured.
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ZONE C • WHAT CAN WE CONCLUDE?

	Random assignment: YES	Random assignment: NO
Random sample: YES	Generalize: yes Cause and effect: yes	Generalize: yes Cause and effect: no
Random sample: NO	Generalize: no Cause and effect: yes	Generalize: no Cause and effect: no

ZONE D • WATCH OUT + BEYOND

Random **selection** lets you generalize to the population.
Random **assignment** supports cause and effect.

★ **BEYOND THE EXAM**
Re-randomize groups many times. If the observed difference is rare, it is evidence against chance alone.

TWO CATEGORICAL VARIABLES

new 2.1–2.2 • old 2.1–2.3

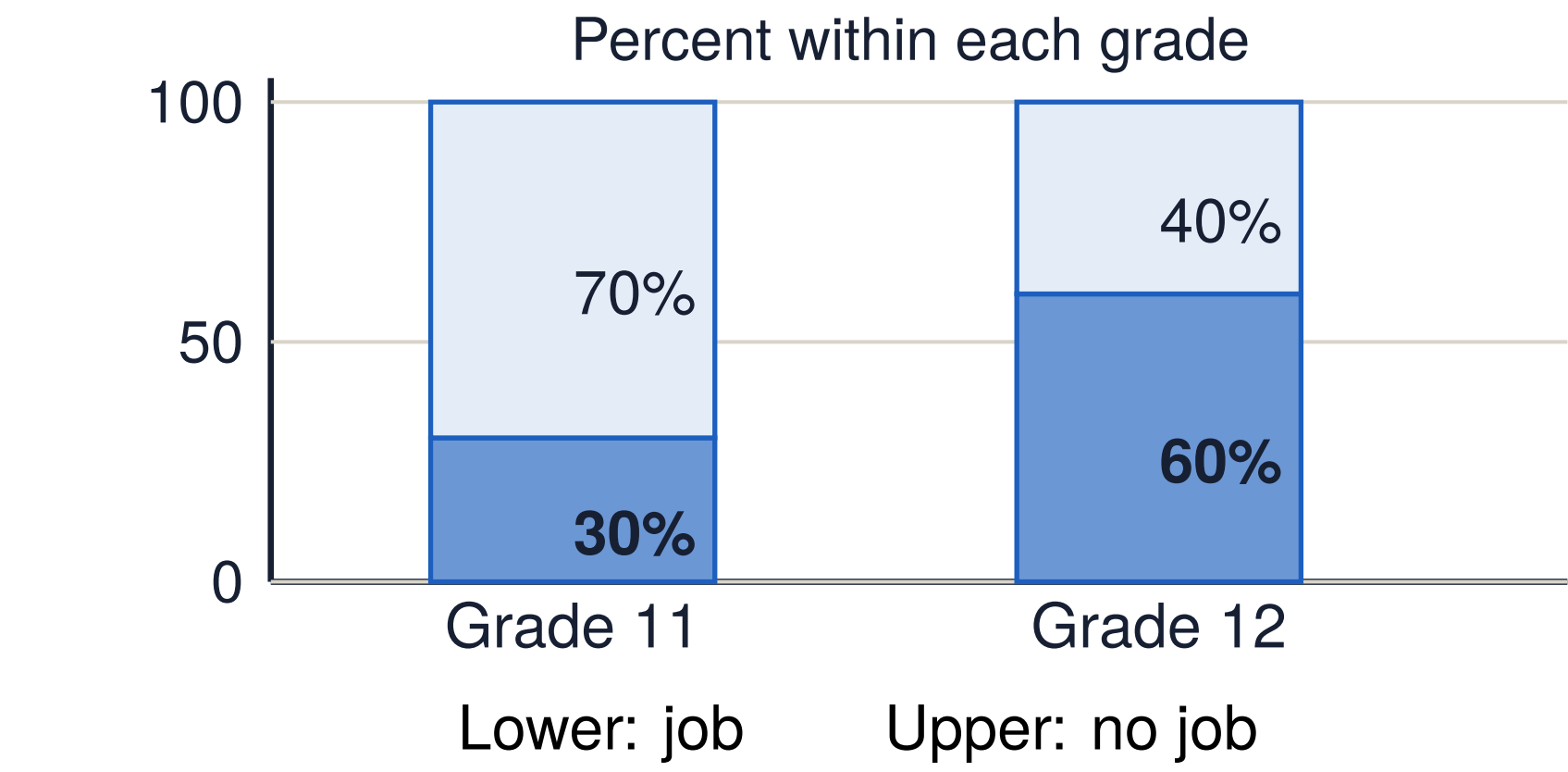
ZONE A • COUNTS, SHARES, AND COMPARISONS

DO STUDENTS HAVE A JOB?

	Grade 11	Grade 12	Total
Job	12	36	48
No job	28	24	52
Total	40	60	100

Within Grade 11: $12/40 = 30\%$ have a job.
Within Grade 12: $36/60 = 60\%$ have a job.

SEGMENTED BARS: EACH GROUP IS 100%



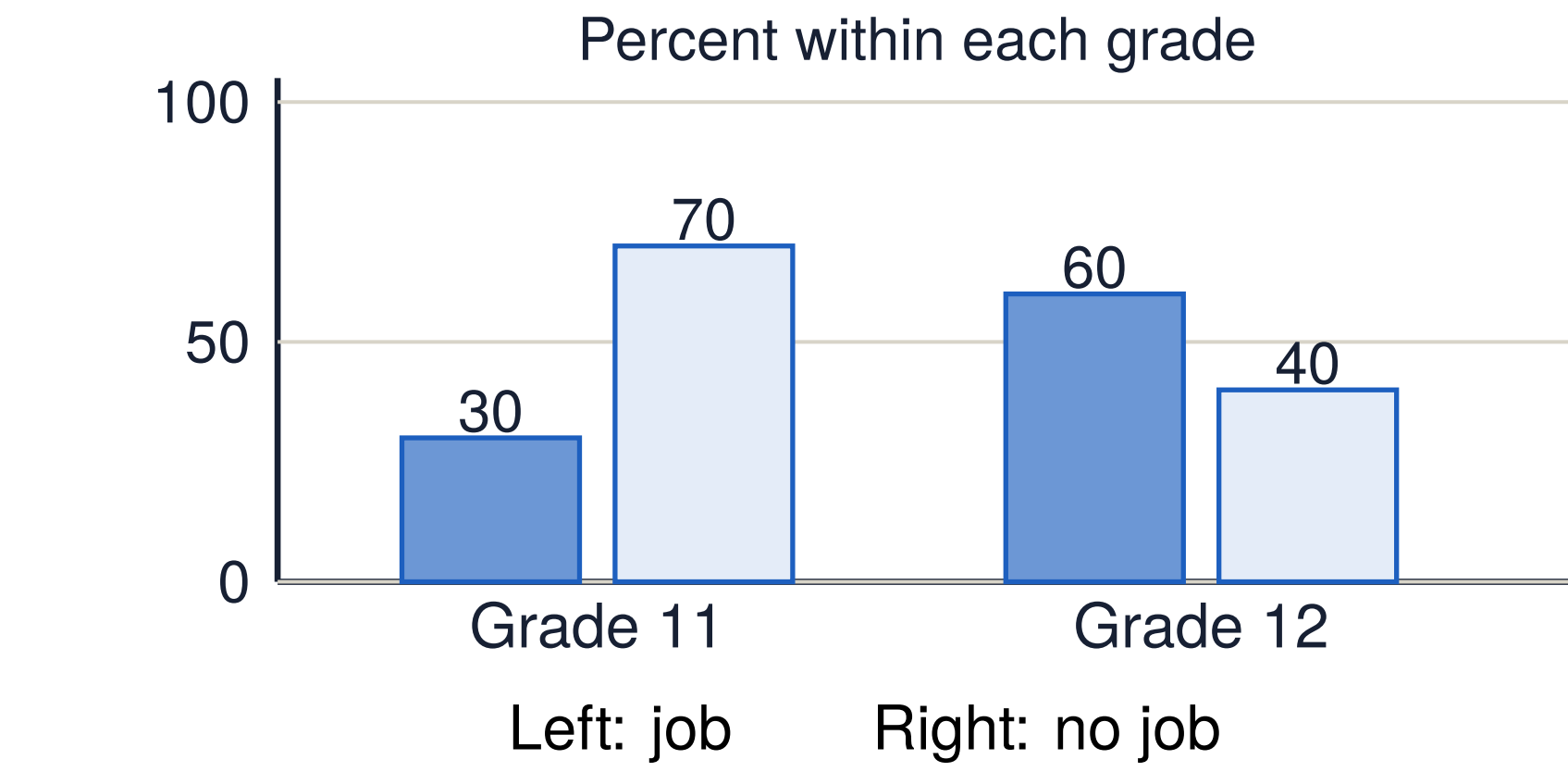
CHOOSE THE DENOMINATOR

$$\text{joint rel. freq.} = \frac{\text{cell}}{\text{grand total}}$$

$$\text{marginal rel. freq.} = \frac{\text{row or column total}}{\text{grand total}}$$

$$\text{conditional rel. freq.} = \frac{\text{cell}}{\text{row or column total}}$$

SIDE-BY-SIDE BARS: COMPARE SHARES



ZONE B • WHAT THE WORDS MEAN

Explanatory variable: defines the groups we compare (grade level).
Response variable: the outcome compared (job status).

CONDITIONAL ON = WITHIN

Restrict the total to that row or column.
Each conditional distribution adds to 100%.

ZONE C • SAY IT IN WORDS

“There is an association because the conditional distribution of _____ differs across _____ groups: _____% versus _____%.”

Name the response, the explanatory groups, and the compared category.

ZONE D • WATCH OUT

COMPARE CONDITIONAL DISTRIBUTIONS
Use shares, not counts, when group sizes differ.
State which group supplies the denominator.

PROBABILITY RULES

new 2.3–2.7 • old 4.1–4.6

ZONE A • THE FORMULAS

$$0 \leq P(A) \leq 1 \qquad P(S) = 1$$

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

mutually exclusive: $P(A \cap B) = 0$

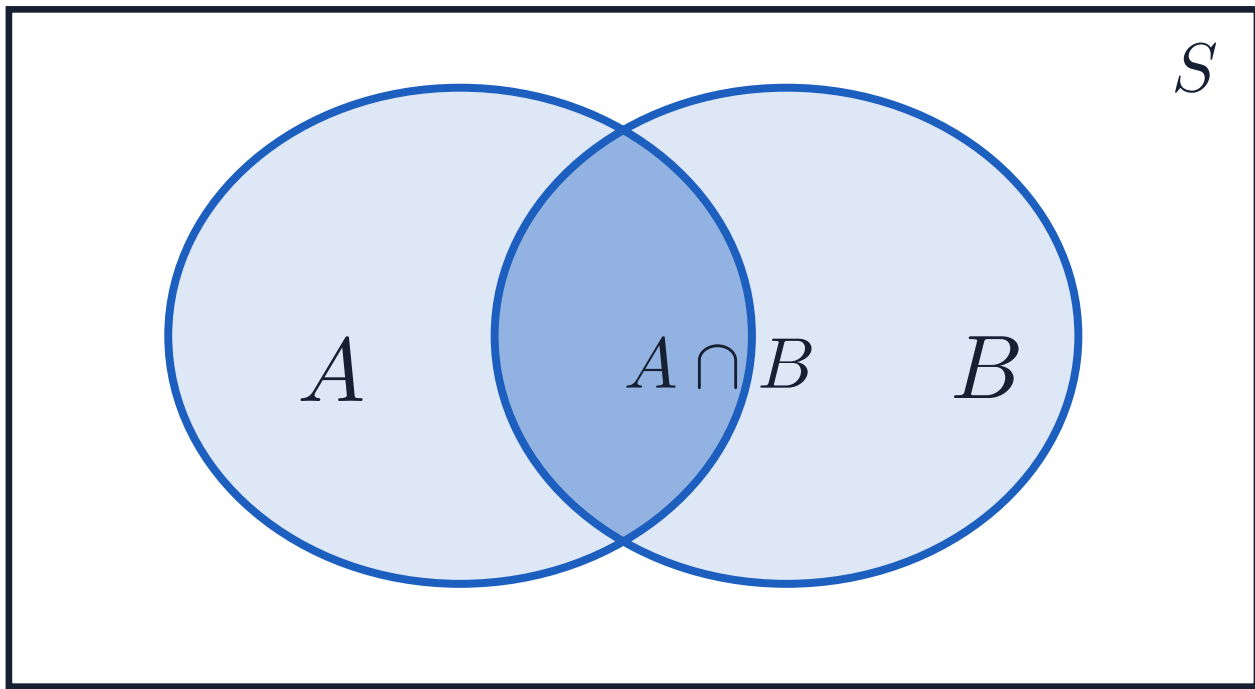
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \qquad P(B) > 0$$

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

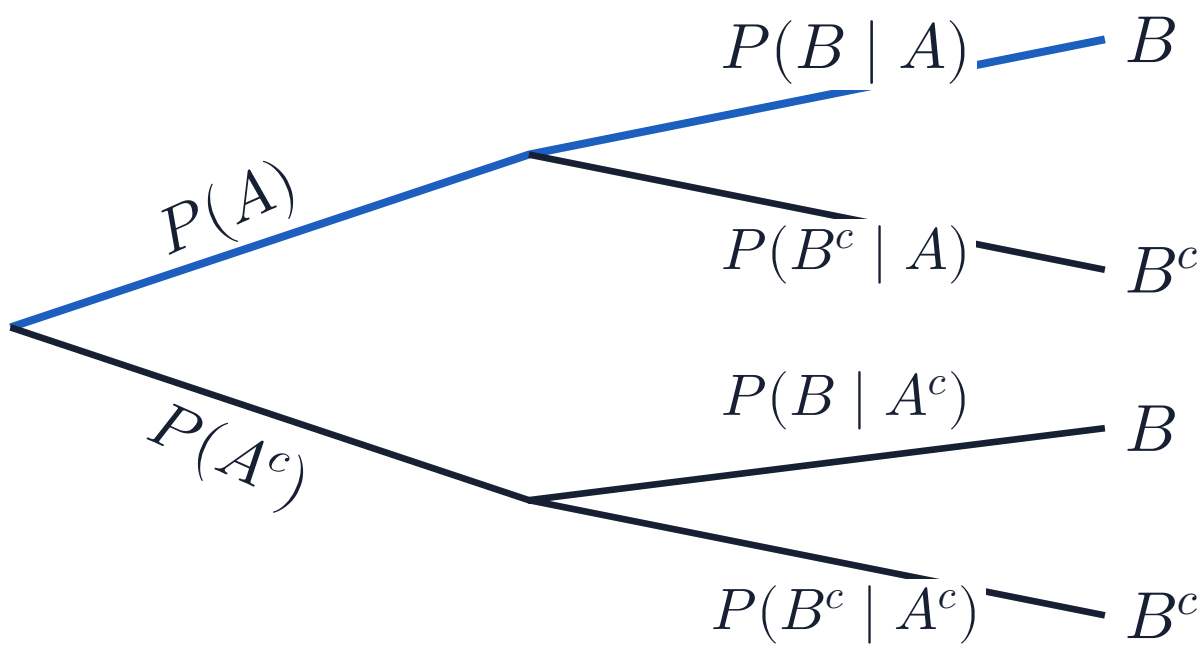
$$\begin{aligned} \text{Independent} &\iff P(A \mid B) = P(A) \\ &\iff P(A \cap B) = P(A) \cdot P(B) \end{aligned}$$

A conditional expression requires a positive probability for its given event.

OR: COUNT THE OVERLAP ONCE



AND: MULTIPLY ALONG A PATH



ZONE B • WHAT THE SYMBOLS MEAN

∪: **or** ∩: **and** |: **given**
 A^c : event A does **not** occur.
Outcome: one result.
Event: a set of outcomes.

Sample space S : all possible outcomes.
Law of large numbers: over many repetitions, an event's relative frequency approaches its probability.

ZONE C • THE SIMULATION RECIPE

1 • DESCRIBE ONE TRIAL

Use random digits or the calculator. Match the event and its chance.

2 • REPEAT

Run many trials. Count the trials meeting the success rule.

3 • ESTIMATE

$$P \approx \frac{\# \text{ successes}}{\# \text{ trials}}$$

State the event in context.

ZONE D • WATCH OUT

DISJOINT IS NOT INDEPENDENT

Mutually exclusive events are **never independent**, unless at least one has probability 0.

CHECK IT

Independence is checked, not assumed.

RANDOM VARIABLES & BINOMIAL

new 2.8–2.10 • old 4.7–4.12 • ★ geometric + combining

ZONE A · THE FORMULAS

ANY DISCRETE RANDOM VARIABLE

$$\mu_X = \sum x_i \cdot P(x_i)$$

$$\sigma_X = \sqrt{\sum (x_i - \mu_X)^2 \cdot P(x_i)}$$

Weight each possible value by its probability.
All probabilities are nonnegative and sum to 1.

SHIFT + SCALE: $Y = a + bX$

$$\begin{aligned}\mu_Y &= a + b\mu_X \\ \sigma_Y &= |b|\sigma_X\end{aligned}$$

BINOMIAL: COUNT THE SUCCESSES

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

$$\begin{aligned}\mu_X &= np \\ \sigma_X &= \sqrt{np(1 - p)}\end{aligned}$$

Binary: success or failure
Independent trials
Number of trials is fixed
Same success probability p

Possible counts: $x = 0, 1, \dots, n$.
Check **BINS** before using the formulas.

ZONE B · WHAT THE SYMBOLS MEAN

X : a numerical outcome of a chance process
 x_i : one possible value
 μ_X : long-run mean σ_X : SD

Binomial X : successes in n trials
 p : success probability on one trial
 $\binom{n}{x}$: number of success/failure orderings

ZONE C · SAY IT IN WORDS

“If we repeated this many times, the average _____ would be about _____.”

Name the variable, the value of μ_X , and its units.

ZONE D · WATCH OUT + BEYOND

COMBINE VARIANCES

For independent variables, add **variances**; then take the square root. Do not add SDs.
Subtracting variables still adds their variances.
Use $|b|$, not b , when scaling an SD.

★ BEYOND THE EXAM

Combining random variables

$\mu_{aX+bY} = a\mu_X + b\mu_Y$
 $\text{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2$
Mean rule: always. Variance rule: **independent** X, Y .

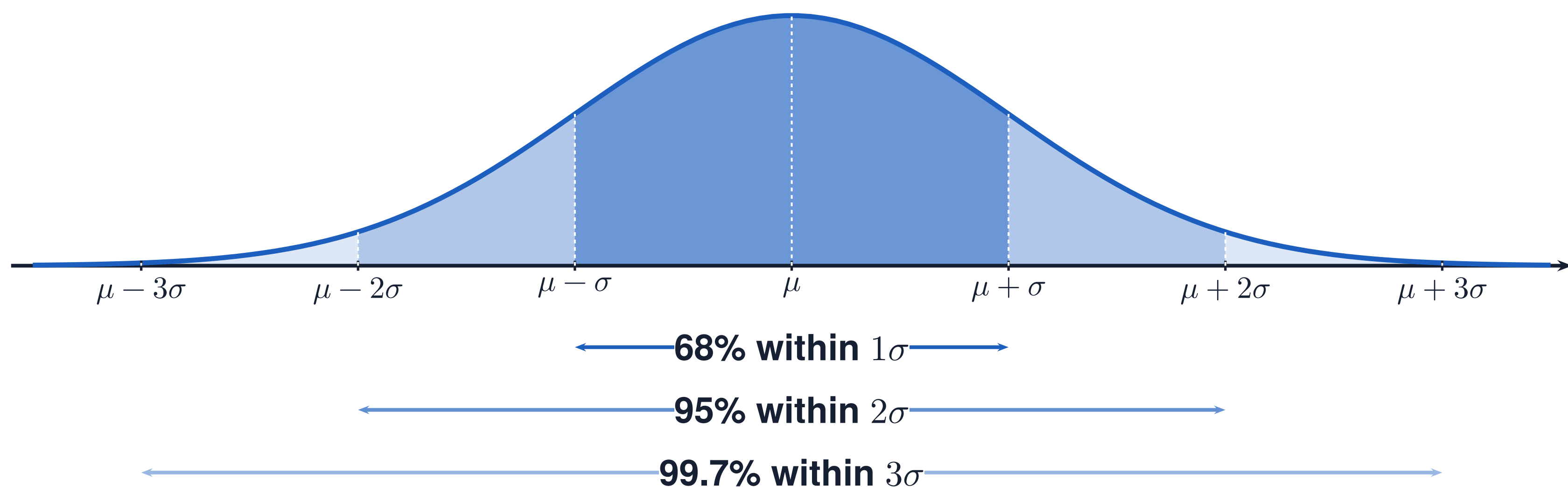
Geometric: trial of the first success

Independent binary trials, same $0 < p \leq 1$; count $x = 1, 2, 3, \dots$
 $P(X = x) = (1 - p)^{x-1} \cdot p$
 $\mu_X = \frac{1}{p} \quad \sigma_X = \frac{\sqrt{1 - p}}{p}$

THE NORMAL DISTRIBUTION

new 2.11 • old 1.10

ZONE A · THE CURVE + THE FORMULA



$$z = \frac{x - \mu}{\sigma}$$

$x \longrightarrow z \longrightarrow \text{area}$

normalcdf: value to probability

$\text{area} \longrightarrow z \longrightarrow x$

invNorm: probability to value

ZONE B · WHAT THE SYMBOLS MEAN

CENTER + SPREAD

$$X \sim N(\mu, \sigma)$$

μ : mean σ : standard deviation

Total area under the curve is 1.

STANDARDIZE THE DISTANCE

z counts SDs from the mean.

Positive: above μ . Negative: below μ .

At the mean, $z = 0$.

ZONE C · SAY IT IN WORDS

“_____ is _____ standard deviations [above / below] the mean of _____.”

“About _____% of _____ have _____ below _____.”

ZONE D · WATCH OUT

DRAW BEFORE YOU CALCULATE

Draw the curve. Mark the value.
Shade the requested region first.

PERCENTILE = AREA TO THE LEFT

Multiply the left-tail area by 100.
The 68–95–99.7 rule gives approximate percentages for a normal model.

SAMPLING DISTRIBUTIONS & THE CLT

new 2.12, 3.1–3.2, 4.1 • old 5.1–5.7

ZONE A · THE FORMULAS + CONDITIONS

1 · RANDOM: START WITH A RANDOM SAMPLE

SAMPLE PROPORTION $\hat{p}$ Population proportion p	SAMPLE MEAN $\bar{x}$ Population mean μ , SD σ
$\mu_{\hat{p}} = p$	$\mu_{\bar{x}} = \mu$
$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$	$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

2 · 10%: $n < 0.10 \cdot N$ when sampling without replacement.
Draws are close enough to independent to use these SD formulas.

3 · LARGE COUNTS

The distribution of $\hat{p}$ is approximately normal when
 $np \geq 10$ and $n(1-p) \geq 10$.

3 · NORMAL / CENTRAL LIMIT THEOREM

Normal population $\Rightarrow$ normal $\bar{x}$, for any n .
Otherwise, CLT gives an **approximate** normal shape
for large n . The $n \geq 30$ guideline is approximate;
strong skew can need a larger sample.

ZONE B · PARAMETERS VS STATISTICS

	Parameter	Statistic
Mean	μ	$\bar{x}$
SD	σ	s
Proportion	p	$\hat{p}$
Slope	β	b

Parameters are Greek or unhatted. Statistics wear a hat or a bar (or use the sample symbols s and b).

Unbiased: the sampling distribution is centered at the parameter.

Variability shrinks with $\sqrt{n}$: **quadrupling n halves the SD.**

ZONE C · SAY IT IN WORDS

“The sampling distribution of _____ is the distribution of that statistic from all possible samples of size $n = \rule{1cm}{0.4pt}$ from _____.”

One sample gives one statistic; repeated samples give a distribution.

ZONE D · WATCH OUT

KEEP n AND N SEPARATE

n is sample size; N is population size.
Sample size controls these SDs.
Use N for the 10% check.

SD VS ESTIMATED SD

$\sigma_{\bar{x}} = \sigma/\sqrt{n}$ needs population SD σ . With sample SD s ,
 $s/\sqrt{n}$ is an **SE** estimating the SD of $\bar{x}$. See P13.

CONFIDENCE INTERVALS

master recipe • also used in Units 4 and 5

ZONE A • THE FORMULAS

$$\text{statistic} \pm \underbrace{(\text{critical value})(\text{standard error})}_{\text{margin of error}}$$

$$ME = z^* \cdot SE \quad (\text{proportions})$$

$$ME = t^* \cdot SE \quad (\text{means})$$

Use the procedure’s *df* to choose *t**.

$$\text{width} \propto \frac{1}{\sqrt{n}} \implies \text{halve the margin} \implies \text{quadruple } n$$

THE FOUR-STEP FRAME

1	STATE	Name the parameter in context.
2	PLAN	Name the interval; check conditions.
3	DO	Formula, substitutions, and interval.
4	CONCLUDE	Interpret the interval in context.

COMMON *z** VALUES

confidence	<i>z</i> *
90%	1.645
95%	1.960
99%	2.576

Get *t** from the table or calculator using **df**.

ZONE B • CONDITIONS — ALWAYS IN THIS ORDER

1 • RANDOM

Random sample or
randomized experiment.

2 • 10%

Sampling without replacement:
n < 0.10*N*.

3 • NORMAL / LARGE COUNTS

CI: *n**p̂* ≥ 10, *n*(1 − *p̂*) ≥ 10
Test: *n**p*₀ ≥ 10, *n*(1 − *p*₀) ≥ 10 (see P11)
means: normal population, *n* ≥ 30, or no strong
skew/outliers

ZONE C • SAY IT IN WORDS

“We are ____% confident that the interval from ____ to ____ captures the true ____.”

Confidence level: If we took many samples and built an interval each time,
about ____% of those intervals would capture the true ____.

ZONE D • WATCH OUT

THE INTERVAL ESTIMATES THE PARAMETER. THE LEVEL DESCRIBES THE METHOD.

Never say the fixed parameter has a 95% chance of being inside the interval. After the interval is calculated,
it either captures the parameter or it does not.

SIGNIFICANCE TESTS

master recipe • also used in Units 4 and 5

ZONE A · THE FORMULAS + THE FOUR-STEP FRAME

$$z = \frac{\text{statistic} - \text{parameter under } H_0}{\text{standard error of the statistic}}$$

SE estimates the statistic’s sampling SD; use the test’s null model. A t -procedure uses t with its **df**.

$$P\text{-value} = P(\text{observed or more extreme statistic} \mid H_0 \text{ true})$$

THE SAME FOUR STEPS AS A CONFIDENCE INTERVAL

- 1

STATE

Define the parameter. Give H_0 , H_a , and α in context.
- 2

PLAN

Name the test. Check Random $\rightarrow$ 10% $\rightarrow$ Normal / Large Counts.
- 3

DO

Statistic, df if needed, and P-value. Draw and shade the curve.
- 4

CONCLUDE

Compare P with α ; state evidence for H_a in context.

THE DECISION

- $P \leq \alpha$

$\rightarrow$ reject H_0

Convincing evidence for H_a .
- $P > \alpha$

$\rightarrow$ fail to reject H_0

Not convincing evidence for H_a .

ZONE B · ERRORS + POWER

DECISION × TRUE STATE

	H_0 true	H_a true
Reject H_0	Type I error	Correct
Fail to reject H_0	Correct	Type II error

α : significance level.

β and power depend on a specific true effect.

$$\begin{aligned}\alpha &= P(\text{Type I error}) \\ &= P(\text{reject } H_0 \mid H_0 \text{ true})\end{aligned}$$

$$\begin{aligned}\beta &= P(\text{Type II error}) \\ &= P(\text{fail to reject } H_0 \mid H_a \text{ true})\end{aligned}$$

Power = $1 - \beta$
Increases with larger n , larger α , or a larger true effect, with other factors held fixed.

ZONE C · SAY IT IN WORDS

“Because the P -value of _____ is [less/greater] than $\alpha =$ _____, we [reject / fail to reject] H_0 . There [is / is not] convincing evidence that _____.”

Fill the last blank with H_a in context. At $P = \alpha$, use “equal to” and reject.

ZONE D · WATCH OUT

KEEP THE DECISION PRECISE

Say “fail to reject H_0 .”
Never “accept H_0 .”

NAME THE PARAMETER

Hypotheses use p or μ ,
not $\hat{p}$ or $\bar{x}$.

CHOOSE TAILS BEFORE DATA

One-sided vs two-sided changes P. Let the question choose the tail.

PROPORTIONS TOOLBOX

new 3.2–3.13 • old 6.1–6.11

ZONE A · ONE SAMPLE / TWO INDEPENDENT SAMPLES

ONE PROPORTION · CONFIDENCE INTERVAL

$$s_{\hat{p}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

ONE PROPORTION · TEST

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

$H_0 : p = p_0$. Use the null value p_0 in the denominator.

TRUE SD · WHEN POPULATION PROPORTIONS ARE KNOWN

$$\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1(1 - p_1)}{n_1} + \frac{p_2(1 - p_2)}{n_2}}$$

Standard error ($s_{\text{statistic}}$, or SE) estimates the true SD of a sampling distribution using sample data.

TWO PROPORTIONS · CONFIDENCE INTERVAL

$$s_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

TWO PROPORTIONS · TEST OF $H_0 : p_1 = p_2$

$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2}$$

$$SE_{\text{pooled}} = \sqrt{\hat{p}_c(1 - \hat{p}_c) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)(1/n_1 + 1/n_2)}}$$

x_i : successes; n_i : sample size; $\hat{p}_i = x_i/n_i$.

ZONE B · CONDITIONS + WHICH PROPORTION TO USE

- 1 · **Random**: random sample or randomized experiment; separate groups independent.
- 2 · **10%**: when sampling without replacement, $n < 0.10N$ in each population.
- 3 · **Large Counts**: at least 10 expected successes and 10 expected failures.

CI: use the sample proportion

$$n\hat{p} \geq 10, \quad n(1 - \hat{p}) \geq 10$$

Two groups: check each $\hat{p}_i$ with its own n_i .

Test: use the null model

$$np_0 \geq 10, \quad n(1 - p_0) \geq 10$$

Two groups: use $\hat{p}_c$; check both $n_i\hat{p}_c$ and $n_i(1 - \hat{p}_c)$.

ZONE C · SAY IT IN WORDS

“We are ____% confident the true proportion of _____ is between _____ and _____.”

For two groups, name the difference $p_1 - p_2$ in context and keep the subtraction order.

ZONE D · WATCH OUT

CI: estimate each unknown proportion with its own $\hat{p}$.

One-sample test: use p_0 . **Two-sample test**: pool under $p_1 = p_2$.

Keep the CI unpooled. Its groups may have different true proportions.

CHI-SQUARE TESTS

new 3.14–3.15 • old 8.4–8.6 • ★ old 8.1

ZONE A · COUNTS, A RIGHT TAIL, AND A TEST SELECTOR

$$\chi^2 = \sum_{df=(r-1)(c-1)} \frac{(O - E)^2}{E}$$

$$E = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$$

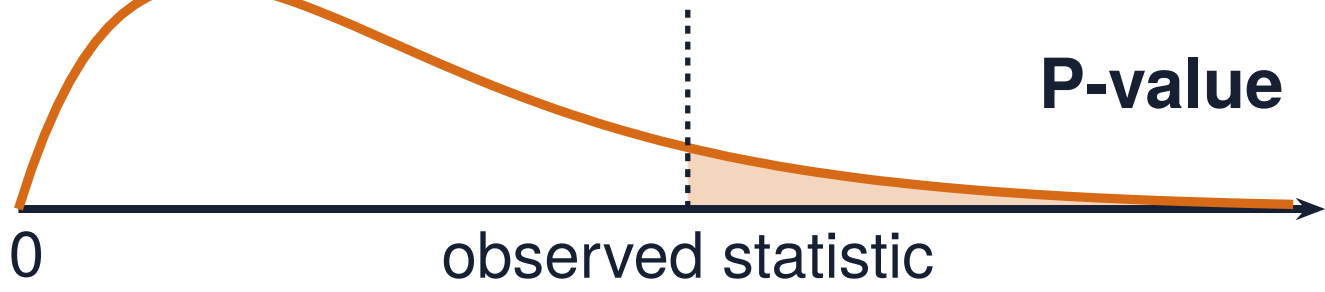
r : number of rows; c : number of columns.
Sum every cell’s contribution.

CHOOSE FROM THE DATA COLLECTION DESIGN

Test	Data shape	H_0 says
Homogeneity	One categorical variable; several populations or treatments	Same distribution in every population or treatment.
Independence	Two categorical variables; one population	The variables are not associated.
★ Goodness of fit	One categorical variable; claimed proportions	Matches the claimed distribution.

H_a : distributions differ; variables are associated; at least one claimed proportion differs, respectively.

SHADE TO THE RIGHT



Large values mean a larger mismatch. All chi-square test P-values are right-tail areas.

ZONE B · CONDITIONS, SYMBOLS, AND OUTPUT

1 · **Random** sample or assignment → 2 · **10%** $n < 0.10N$ without replacement
→ 3 · **Large Counts: every expected count** $E \geq 5$. Check each sampled population.

OBSERVED VS EXPECTED

O : observed count. E : count expected under H_0 .

$$\text{standardized residual} = \frac{O - E}{\sqrt{E}}$$

two-way test $df=(r-1)(c-1)$

Positive: excess; negative: shortfall.

COMPUTER-OUTPUT DECODER

$\chi^2 = 9.21$, $df = 2$, $P = .010$
9.21: total standardized discrepancy.
2: selects the null chi-square curve.
.010: under H_0 , chance of a statistic this large or larger.

ZONE C · SAY IT IN WORDS

“Because the P-value of ____ is [less/greater] than $\alpha =$ ____, we [reject/fail to reject] H_0 . There [is/is not] convincing evidence of an association between _____ and _____ for _____.”

For homogeneity or GOF, replace “association” with the matching H_a in context.

ZONE D · WATCH OUT + BEYOND

Check expected counts against 5.
Test: overall evidence. Residual: which cell.
Large |residual| marks a stronger departure; its square is the cell’s contribution.

★ **BEYOND THE EXAM**
GOF: $E = np_0$ for each category
 $df = k - 1$; k = number of categories.

MEANS TOOLBOX

t procedures • new 4.1–4.10 • old 7.1–7.10

ZONE A · ONE MEAN, PAIRED DIFFERENCES, TWO MEANS

ONE MEAN

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}} \quad df = n - 1$$

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad df = n - 1$$

$H_0 : \mu = \mu_0$. Standard error $s_{\bar{x}}$ estimates $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ using sample SD s .

TWO MEANS · INDEPENDENT GROUPS

$$s_{\bar{x}_1 - \bar{x}_2} = \sqrt{s_1^2/n_1 + s_2^2/n_2}$$

Left: SE estimates the sampling SD. Right: true SD when population SDs σ_1, σ_2 are known.

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{s_1^2/n_1 + s_2^2/n_2}$$

df : calculator, or $\min(n_1-1, n_2-1)$

PAIRED · WORK WITH THE DIFFERENCES

Same subjects measured twice, or natural pairs: subtract in one consistent order.

$\bar{d}$: mean difference; s_d : SD of differences; n : number of pairs.

$$t = \frac{\bar{d} - \mu_{d0}}{s_d/\sqrt{n}} \quad df = n - 1$$

Usually $H_0 : \mu_d = 0$. Treat the differences as **one sample**; use their shape and their SD.

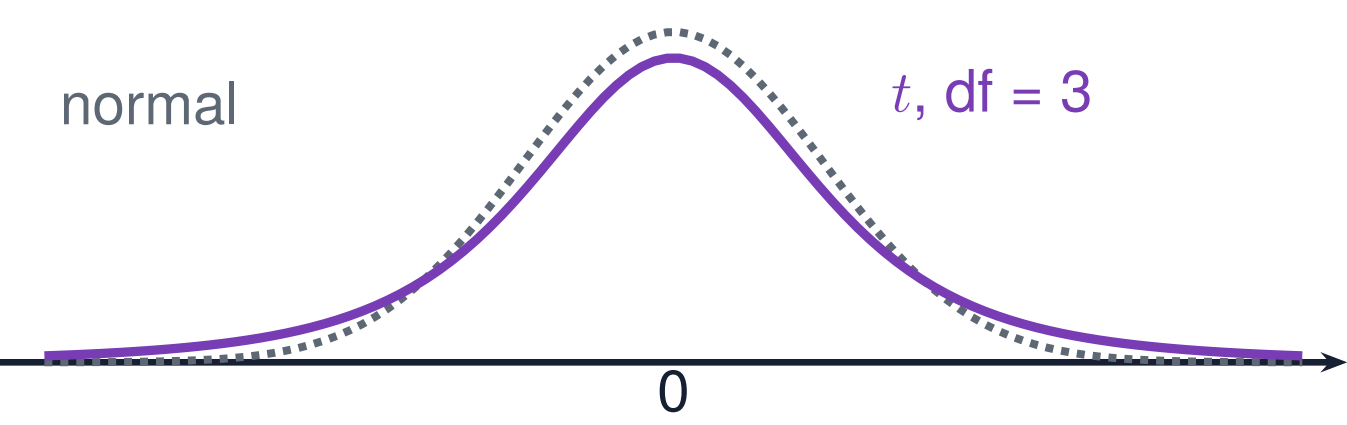
$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

df : calculator, or $\min(n_1-1, n_2-1)$

Test above assumes $H_0 : \mu_1 = \mu_2$. Calculator df is Welch's approximation; the minimum is conservative.

ZONE B · WHY t? + CONDITIONS IN ORDER



t because we replaced σ with s .
Heavier tails; more $df \rightarrow$ closer to normal.

- 1 · **Random:** random sample or randomized experiment.
 - 2 · **10%:** $n < 0.10N$ when sampling without replacement.
 - 3 · **Normal / Large Sample:** normal population, $n \geq 30$, or graph shows no strong skew or outliers.
- Paired: check the **differences**.
Two samples: independent groups; check **each group**.

ZONE C · SAY IT IN WORDS

"We are ____% confident the true mean _____ is between _____ and _____ [units]."

For paired data, name the mean difference μ_d ; for independent groups, name $\mu_1 - \mu_2$.

ZONE D · WATCH OUT

Paired or two-sample? Let the study design decide. Repeated measurements or natural pairs use differences; separate independent groups use two-sample t.

Keep group variances separate for two-sample t. Use the unpooled SE above.

SCATTERPLOTS, CORRELATION & LSRL

new 5.1–5.3, 5.5 • old 2.4–2.6, 2.8

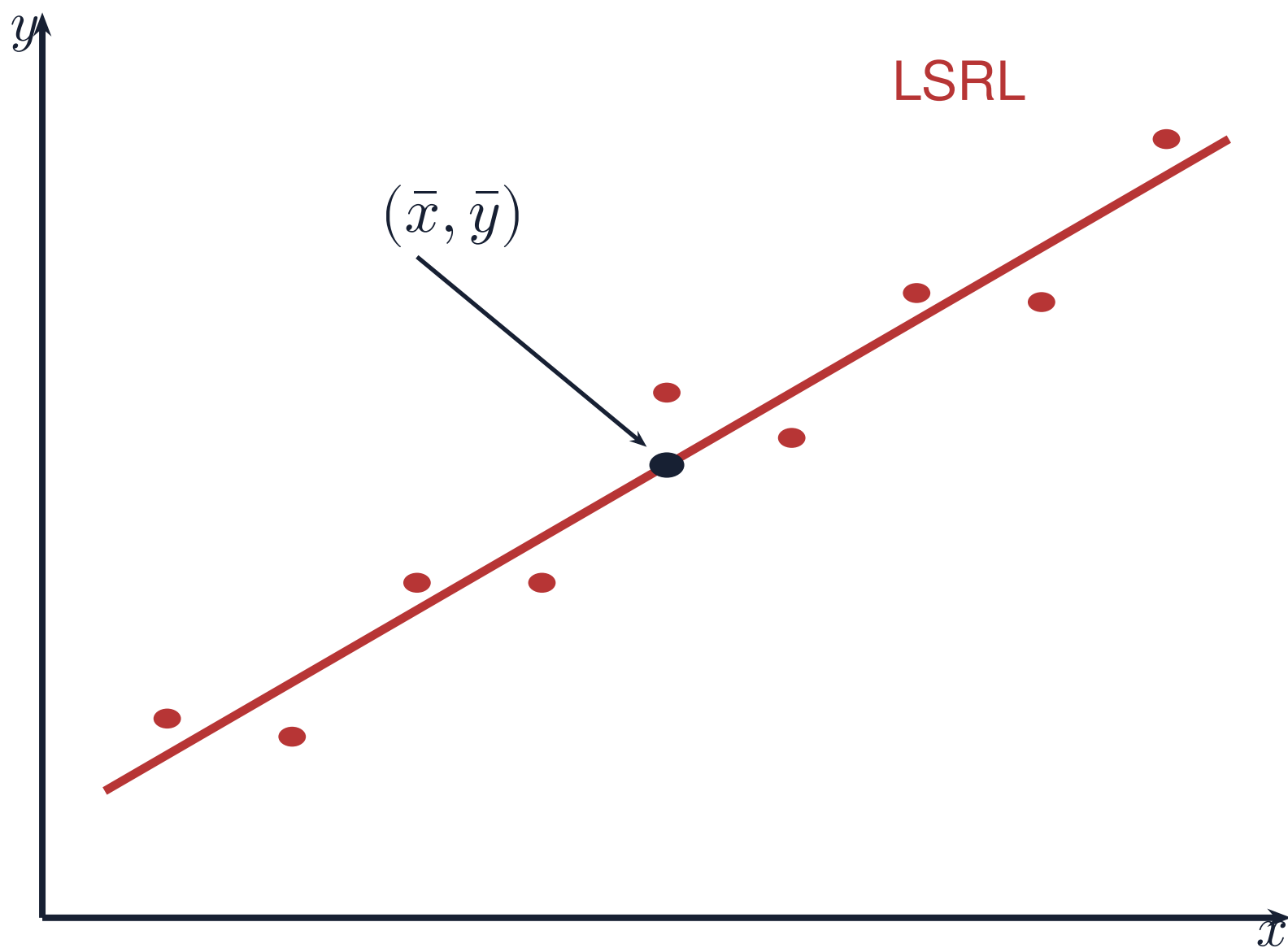
ZONE A · THE LEAST-SQUARES REGRESSION LINE

$$\hat{y} = a + bx$$

$$b = r \frac{s_y}{s_x} \qquad a = \bar{y} - b\bar{x}$$

$$\bar{y} = a + b\bar{x}$$

The LSRL always passes through $(\bar{x}, \bar{y})$. It minimizes the sum of squared vertical residuals.



$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

DIRECTION · FORM · STRENGTH · UNUSUAL POINTS

Describe the association using all four, with the variables and their context.

ZONE B · WHAT THE SYMBOLS MEAN + HOW r BEHAVES

x : explanatory variable
 y : observed response
 $\hat{y}$: **predicted** response
 a : intercept; b : slope
 s_x, s_y : sample SDs; n : number of pairs

$-1 \leq r \leq 1$; **r has no units.**
Swapping x and y leaves r unchanged.
Changing measurement units leaves r unchanged.
 r measures **linear** direction and strength only.
Outliers can change r substantially.

ZONE C · SAY IT IN WORDS

SLOPE

“For each additional _____ [unit of x], the predicted _____ [y] [increases/decreases] by _____ [$|b|$] _____ [units of y].”

INTERCEPT

“When _____ [x] is 0, the predicted _____ [y] is _____ [a] _____ [units].”

This may not make sense in context.

ZONE D · WATCH OUT

Correlation does not establish causation.
Inspect the scatterplot and residual plot (P15) for curvature, even when $|r|$ is large.
Slope has units of y per unit of x ; r has no units.

RESIDUALS, r^2 & MODEL FIT

new 5.4–5.5 • old 2.7–2.8 • ★ old 2.9, 9.1–9.5

ZONE A · THE FORMULAS + THE PLOTS

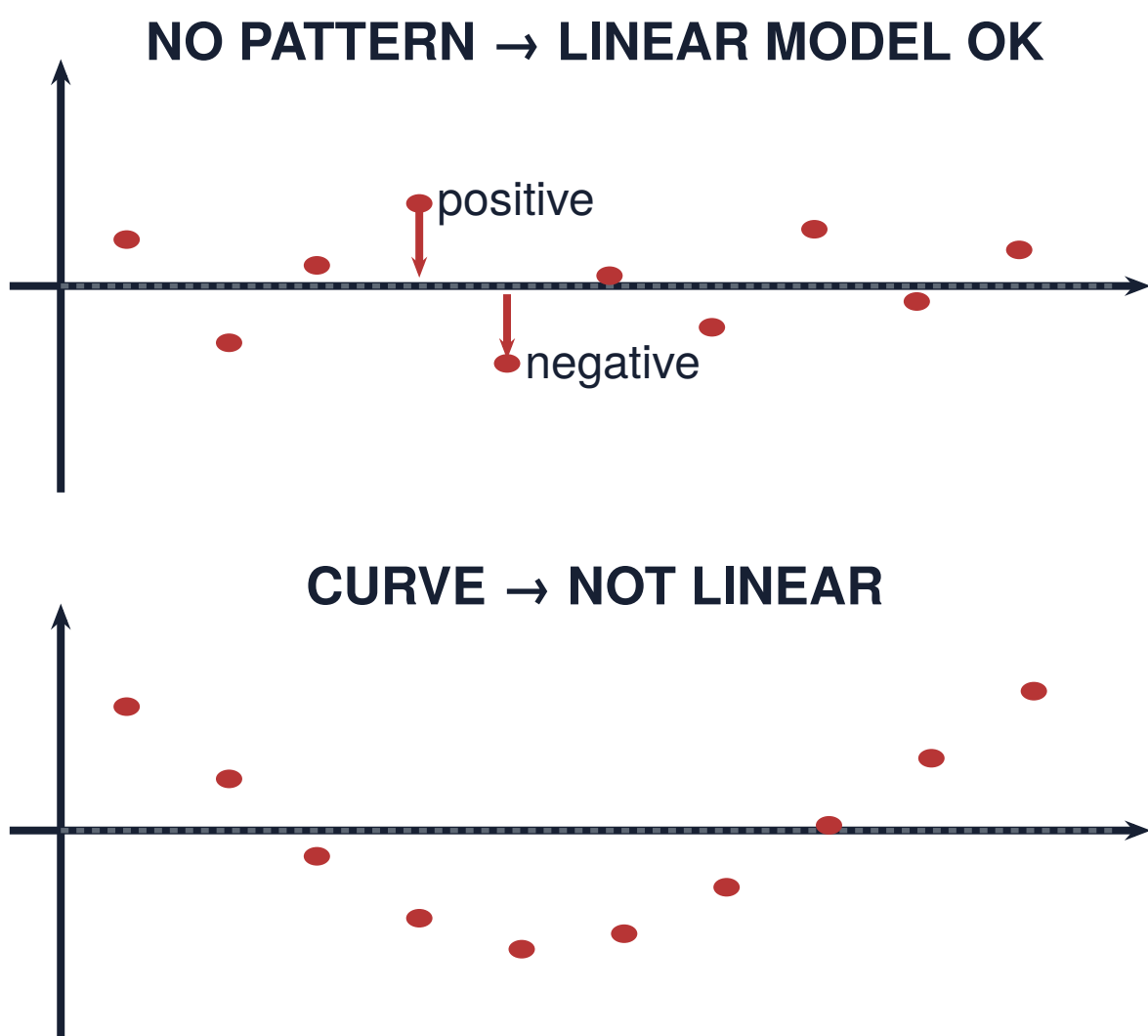
residual = $y - \hat{y}$ =
actual – predicted

r^2 = fraction of the variation in y explained by
the linear relationship with x

$$s = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n - 2}}$$

s is the **typical size of a residual**. It has the same units as
the response variable.

READ THE RESIDUAL PLOT



ZONE B · COMPUTER-OUTPUT DECODER

Predictor	Coef	SE Coef	T	P	Read it as
Constant	12.41	1.86	6.67	<.001	→ a , the intercept
Study hours	4.83	0.72	6.71	<.001	→ b , s_b , t , P -value
$S = 3.17$		→ s	$R\text{-Sq} = 78.9\%$		

ZONE C · SAY IT IN WORDS

“About ____% of the variation in ____ is
explained by the linear relationship with ____.”

“The actual ____ was ____ units ____ what
the line predicted.”

ZONE D · WATCH OUT + ● BEYOND

MODEL-FIT RULES

No extrapolation: do not predict beyond the
observed x -range.

r^2 gives the explained fraction, not the direc-
tion. Use r for direction.

★ BEYOND THE EXAM

Departures: outlier · high leverage · influential point
 $\log(\hat{y}) = a + bx \Rightarrow \hat{y} = 10^{a+bx}$
Slope inference ($df = n - 2$):
 $\mu_b = \beta \quad \sigma_b = \frac{s}{\sigma_x \sqrt{n}}$
 $s_b = \frac{s}{s_x \sqrt{n - 1}} \quad t = \frac{b}{s_b} \quad b \pm t^* s_b$